

HARSHAL JAISINGPURE

+91 9307846620 | hjaisingpure77@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

Government College of Engineering, Yavatmal B.Tech –Electronics &Telecommunication Engineering CGPA: 7.2 / 10	2022– 2026
Smt. Nankibai Wadhwani Junior Science College, Yavatmal HSC, Maharashtra state board Science stream(PCMB) Score: 75.69%	2018–2020
Maharshi Vidya Mandir Yavatmal HSC CBSE Board Score: 81.62%	2012–2018.

TECHNICAL SKILLS

Languages: Python, HTML, CSS, Arduino C++, Kotlin, SQL
Frameworks: Numpy, Flask
Cloud & DB: MySQL, SQLite, Google Cloud Platform
Hardware: Raspberry Pi, ESP32, Aduino bords, 8051uc
Tools: GitHub, VS Code, Arduino IDE, Android studio
AI / ML: Computer Vision, Generative AI

PROJECTS

Wheelio – IoT-Enabled, Android App Controlled, Autonomous Wheelchair Navigator 2026

Tech Stack: *Android, Kotlin, ESP32, Google Maps API, OSRM, Embedded Systems.*

- Developed a full-stack navigation system connecting an Android application to an ESP32-controlled wheelchair via Bluetooth SPP.
- **Manual control mode:** Virtual Joystick control and Maps based autonomous navigation: Implemented 8-directional manual and autonomous navigation, involving route planning, real-time sensor feedback, and GPS tracking.
- **Self driving mode:** Implemented Closed-Loop Control using smartphone magnetometer data to correct for physical motor drift, reducing orientation errors.
- **Obstacle avoidance:** Implemented obstacle detection and avoidance system with ultrasonic sensors(HC-SR04) and LiDAR sensors for a safe user experience.

Demonstrated practical application of AI and embedded systems in assistive healthcare technology.

Object Detection using YOLOv3 and OpenCV

2025

Tech Stack: *Python, OpenCV, YOLOv3, ESP32, COCO dataset*

- Designed and implemented a real-time object detection system leveraging YOLOv3 and OpenCV's DNN module.
- Streamed image data from an ESP32-CAM over HTTP and processed frames for multi-class detection using the COCO dataset.
- Focused on optimizing inference latency and improving detection efficiency under hardware and bandwidth constraints.

- Designed for applications in **assistive healthcare systems**, such as obstacle detection for smart mobility devices, demonstrating practical deployment of deep learning models in resource-constrained environments.
- Designed for **low-cost surveillance scenarios**, enabling real-time detection of objects such as persons and vehicles, with potential applications in **intrusion detection, smart monitoring systems, and edge-based security deployments**.

TRAINING & CERTIFICATIONS

Google Gen AI APAC Edition – Cohort 2

Ongoing

- Actively participating in Google's structured Generative AI learning programme, covering LLMs, prompt engineering, and building AI-powered applications.
- Gaining hands-on experience with Google AI APIs, Google skills lab, Google AI studio and responsible AI development practices.

Certification- Python for Data science, AI and Development by IBM, Coursera

2026

- Developed a strong foundation in Python programming, including core concepts such as data structures, control flow, functions, and object-oriented programming.
- Gained hands-on experience with data analysis libraries (NumPy, Pandas), Jupyter Notebooks, and working with web data through REST APIs and web scraping using BeautifulSoup.

Internship - Embedded systems with PIC microcontroller

Aug 2024 – Sept 2024

- Programmed embedded systems in C using PIC16F877A, 8051 Microcontroller, ESP32, Arduino Uno, and NodeMCU.
- Simulated and tested circuits using Proteus Design Suite; developed firmware with MPLAB X IDE and Keil uVision.
- Built embedded applications involving GPIO, timers, interrupts, and serial communication (UART, I2C, SPI), along with basic PCB design.

ACHIEVEMENTS

- **Winner** – Project Competition, Government College of Engineering Yavatmal (GCOEY), for the Self-Driving Wheelchair (2026).
- Selected for Google Gen AI APAC Edition Cohort 2, a competitive programme for AI enthusiasts across the Asia-Pacific region.
- **Volunteer**- NSS, Techfusion2024(GCOEY), Technovision2025(GCOEY)